

KUNGULWAR SHASHIKANTH

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Education

Indian Institute of Technology Kanpur (IITK) <i>M.Tech, Computer Science and Engineering CPI 8.95/10</i>	Kanpur, India 2025 – Present
Vardhaman College of Engineering <i>B.Tech, Computer Science and Engineering CGPA 8.8/10</i>	Hyderabad, India 2020 – 2024
Sri Chaitanya Junior College <i>Intermediate (Class XII) 948/1000</i>	Hyderabad, India 2018 – 2020
SRK High School <i>Class X (SSC) GPA 9.0/10</i>	Siddipet, India 2018

Relevant Coursework

- Design & Analysis of Algorithms
- Deep Learning for Comp Vision
- Parallel Computing
- Intro to Machine Learning
- Generative Artificial Intelligence
- Algorithmic Information Theory

Research Experience

Indian Institute of Technology Kanpur (IITK) <i>M.Tech Thesis Advisor: Prof. Arnab Bhattacharya</i>	Kanpur, India May 2026 – Present
Ambiguity in Coreference Resolution for Large Language Models	
• Developing a Hindi-language benchmark containing ambiguous, non-ambiguous, and cue-dependent coreference instances to evaluate LLM reasoning over discourse context. • Designed a multiple-choice evaluation framework to measure LLM ability to resolve ambiguous references using contextual information, coreference chains, and discourse cues. • Evaluating how explicit coreference and discourse information affects LLM accuracy in resolving ambiguous references across varying contextual conditions.	

Technical Skills

Languages: Python, C, C++, SQL, JavaScript, Java, Kotlin
Machine Learning: PyTorch, Hugging Face Transformers, TensorFlow Lite, ONNX Runtime, SentenceTransformers, NumPy, Pandas; quantization, ablation studies, active learning
LLM Systems: Retrieval-Augmented Generation (RAG), Prompt Engineering, Structured-output validation, Ollama, Groq
Parallel & HPC: MPI (non-blocking communication, derived datatypes), SLURM, scaling analysis, performance benchmarking
Backend: FastAPI, Node.js, Express, REST APIs, JWT (refresh-token rotation), OAuth 2.0, rate limiting, input sanitization
Frontend / Mobile: React, Vite, React Native (CLI), Redux Toolkit, Monaco, Material Design 3, HTML, CSS3
Databases / Storage: MongoDB (Mongoose), MySQL, SQLite, Redis, Firebase Firestore, MMKV
DevOps & Tools: Docker, Docker Compose, nginx, Git, GitHub, Postman, Vercel, Render, Gradio, Hugging Face Hub

Projects

- SmartSMS: On-Device SMS Manager (Android) | Self Project | Kotlin, TFLite, Node.js**
- Fine-tuned DistilBERT for 6-class SMS classification, deployed on-device via TFLite with a custom Kotlin WordPiece tokenizer: **100% OTP recall**, no user data leaving the device.
 - Benchmarked DistilBERT against MobileBERT; ablation showed removing hard negatives dropped adversarial accuracy from **100% to ~47%**.
 - Diagnosed INT8 quantization accuracy loss and shipped FP32 after evaluation; built an active-learning pipeline for continuous retraining.
 - Built the Node.js/Express backend: JWT with refresh-token rotation, Google OAuth, multi-device sync, Docker Compose, rate limiting and input sanitization.

- Developed the React Native (CLI) Android app with custom Kotlin native modules, MMKV storage, WorkManager background processing, and default-SMS-role support.

CodeSense AI: Local-First Code Review Assistant | *Self Project* | *FastAPI, Qwen2.5-Coder, Ollama*

- Built a code review tool running inference against a local LLM (qwen2.5-coder:7b via Ollama), exposing style review, complexity analysis, and edge-case generation over FastAPI with a React/Vite/Monaco frontend, SQLite persistence, and Docker Compose.
- Designed a swappable provider abstraction so adding a cloud backend (Groq) required one new file and one factory entry — no change to routes, services, schemas, or frontend; enforced the boundary with an architectural fitness test that fails the build if httpx is imported outside providers/.
- Hardened structured LLM output: mandatory-evidence response schemas, a validator that drops self-refuting claims, JSON parse-and-retry, and temperature pinned at 0.1 after default sampling produced scores of 89/70/30 on identical input.
- Benchmarked latency across models and deployments: **2.2–6.3 s** local (RTX 5060, n=3/cell) versus **0.7–1.9 s** server-side on Groq, separating network from inference; documented that a 7B model self-contradicted to $O(n^2)$ on `sum(range(n))` while a 120B model was correct on all 5 runs with zero JSON retries.

RescueSeg: Hybrid-Attention Transformer for Damage Segmentation | *Team Project* | *PyTorch*

- Co-developed a hybrid encoder combining Overlapping Cross Attention and Global Sparse Attention through a learnable fusion gate, with FPN, Attention U-Net decoder, and boundary refinement.
- Achieved **77.35% mIoU** at **44 ms** inference on RescueNet (11 classes), outperforming a Segmenter baseline under identical settings.
- Ran a 3-variant ablation showing boundary refinement and auxiliary supervision prevent NaN divergence (epochs 14–20), improving optimization stability.

Music Genre Classification from Audio Spectrograms | *Course Assignment, CS776 — 7 members* | *Python, PyTorch*

- Implemented CNN primitives from scratch instead of built-in layers: 2D convolution via `unfold` sliding windows and an `einsum` contraction, plus max pooling, channel normalization, and the classifier head.
- Built a ResNet-v2 pre-activation bottleneck network (**380,920 parameters**) over 3-channel inputs stacking log-mel spectrogram, MFCC, and chroma features per clip.
- Scored **0.92993 macro F1** on the competition leaderboard across **16 classes** (14,550 held-out test samples); validation macro F1 differed by only 0.0002.
- Owned training methodology: optimizer comparison (SGD / AdamW / Adam), batch-size generalization analysis, **OneCycleLR** scheduling, label smoothing, early stopping.
- Ran ablations on input representation, resolution ($124 \times 92 \rightarrow 248 \times 184$), and per-feature normalization.

Text-to-SQL via Dynamic Few-Shot Prompting | *Course Project, 3 members* | *LLMs, RAG, SentenceTransformers*

- Advised by Prof. Arnab Bhattacharya and Prof. Subhajit Roy. Built a **retrieval-augmented** pipeline that embeds each question with a SentenceTransformer (all-MiniLM-L6-v2) and retrieves the top-K most cosine-similar solved examples from a held-out knowledge base to construct the prompt.
- Ran a controlled comparison of four prompting strategies across two models (Gemini 2.5 Flash, Llama 3.1 8B) and two benchmarks (Spider, BIRD), evaluated by execution accuracy on a 200-sample subset per dataset.
- Raised Gemini execution accuracy on **BIRD from 33.0% to 52.5%** and on **Spider from 61.0% to 74.5%** over the zero-shot baseline; Llama improved from 8.0% to 32.5% on BIRD.
- Swept retrieval depth $K = 1-5$, finding gains saturate near $K = 3-4$ and attributing the ceiling to context saturation and BIRD's implicit foreign keys and non-standard column naming.

Distributed 3D Stencil Solver with MPI Halo Exchange | *Course Assignment, CS633 — 5 members* | *C, MPI, HPC*

- Built a distributed d -point stencil solver over a $p_x \times p_y \times p_z$ Cartesian process grid with six-neighbour ghost-cell halo exchange.
- Eliminated pack/unpack overhead by communicating non-contiguous halo faces through MPI derived datatypes.
- Overlapped communication with computation via non-blocking sends/receives, computing interior points during message transit; replaced array copies with pointer swapping.
- Benchmarked on **PARAM Rudra** across **32–96 processes**, measuring a consistent **1.17–1.20 $\times$ speedup** at 240^3 with communication overhead limiting gains at smaller sizes.

Animerch: E-Commerce Web Application | *Self Project* | *React, Redux Toolkit*

- Built an e-commerce storefront with cart, product listing, and Redux Toolkit state management. (*Full-Stack derivative only.*)